



RESEARCH REPORT

 **UK Quantum Computing
Companies: Detailed Industry Analysis**

20 Company Deep Profiles · Hardware, Software, Security & Defense

March 2026 · Prepared by Paul Oh

www.paulncompanies.com

£670M Govt QC Commitment (10yr)	75+ Active QC Companies	\$10B Quantinuum Valuation	2029 Apollo Fault-Tolerant Target
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Executive Summary

The United Kingdom is the world's fourth-largest quantum computing ecosystem and Europe's most commercially mature, punching decisively above its weight relative to its size. With Quantinuum—the world's highest-fidelity quantum computer company—headquartered in Cambridge, a Nobel-calibre research base spanning Oxford, Cambridge, Bristol, and Imperial College, and a National Quantum Technologies Programme that has invested over £1 billion since 2014, the UK has built a quantum ecosystem that leads Europe on commercial deployment and exit quality.

The UK's most defining quantum event of 2025 was the acquisition of Oxford Ionics by IonQ for approximately \$1.075 billion—the largest quantum exit in UK history and one of the largest in the global quantum industry. This validated the UK's spinout-to-acquisition pipeline and demonstrated that British quantum hardware can compete at the highest levels of the global market. Meanwhile, Quantinuum filed a confidential S-1 for what could be the largest quantum IPO ever, with target valuations reported at \$20 billion or more.

Beyond the flagship companies, the UK quantum ecosystem has matured across every vertical: Riverlane leads globally in quantum error correction (QEC) middleware with 60%+ of quantum hardware companies as partners; ORCA Computing has built the world's first rack-mountable room-temperature photonic quantum computer; Nu Quantum raised \$60 million to build Europe's first industrial quantum networking laboratory; PQShield became the go-to hardware PQC provider for IoT and automotive; and KETS Quantum Security is deploying integrated photonics QKD in real fibre networks.

The UK government committed £670 million over 10 years in June 2025—part of a broader £1 billion+ frontier technology package—in direct response to concerns about foreign acquisition of UK quantum assets. This investment, combined with £76 million for National Compute Resources (UKRI, March 2026) and the National Quantum Computing Centre (NQCC) at Harwell, creates the sovereign infrastructure layer that UK quantum companies need to develop and deploy.

For Korea, the UK quantum ecosystem presents a high-value partnership opportunity defined by complementarity rather than competition with US and French quantum strategies. Quantinuum's world-leading gate fidelity, Riverlane's hardware-agnostic QEC stack, ORCA's room-temperature deployability, and PQShield's NIST-certified PQC silicon all address specific Korean gaps. The UK–Korea science and innovation partnership, AUKUS alignment, and Cambridge–Seoul academic relationships provide the bilateral framework.

UK Quantum Ecosystem: Company Overview

The United Kingdom hosts over 75 active quantum technology companies across hardware, software, security, sensing, and networking—the largest and most commercially developed quantum ecosystem in Europe. 20 companies are profiled in detail (★).

Quantum Hardware: Computing

Company	Technology	Stage	Key Highlight
Quantinuum ★	Trapped-ion H-Series (QCCD) + full-stack software	Private (\$10B val)	Helios 99.9975% fidelity · DARPA QBI · S-1 filed IPO
OQC ★	Superconducting (Coaxmon/Toshiko) 3D transmon	Private (\$100M+)	NY Digital Realty deploy · 500Q wafer-scale 2026 · Genesis
ORCA Computing ★	Photonic, room-temperature, telecom-standard photons	Private (\$25M)	PT-3 commercial 2026 · Quantum Data Centre of the Future
Universal Quantum ★	Trapped-ion microwave (no lasers), modular chip arch	Private (£100M+)	Sussex spinout · first full microwave ion trap system
Quantum Motion ★	Silicon CMOS qubits, CMOS-compatible manufacturing	Private (£62M+)	Bosch Ventures · DARPA QBI · UCL EPSRC SiQEC
Oxford Ionics ★	Ion-trap-on-a-chip (acquired by IonQ \$1.075B June 2025)	Acquired	Largest UK quantum exit ever · 2M qubit roadmap to 2030
SEEQC UK	Digital quantum computing (superconducting SFQ circuits)	Private (US/EU HQ)	DARPA QEC testbed · full-stack digital quantum processor

Quantum Error Correction & Software

Company	Technology	Stage	Key Highlight
Riverlane ★	QEC middleware (Deltaflow stack) — hardware-agnostic	Private (\$75M S-C)	60%+ of QC companies as partners · Rigetti/IQM/NQCC
Phasecraft ★	Quantum algorithms (materials science, pharma)	Private (£10M+)	UCL/Bristol spinout · Google · BASF · AstraZeneca
QuantroIOx ★	ML-based automated quantum control software	Private (£3M+)	Oxford spinout · works with all major QC hardware
Parity QC	Parity quantum computing architecture (EU/UK)	Private	Novel error correction architecture from Innsbruck/UK
Quantum Dice	QRNG hardware (self-certifying source-device-independent)	Private (£5M+)	Oxford spinout · automotive · defence · finance

Quantum Networking & Communications

Company	Technology	Stage	Key Highlight
Nu Quantum ★	Quantum networking hardware (entanglement distribution)	Private (\$60M S-A)	Dec 2025 \$60M · Europe's first industrial trapped-ion networking lab

Company	Technology	Stage	Key Highlight
KETS Quantum Sec ★	Integrated photonics QKD hardware, Bristol	Private (£10M+)	Quantum Data Centre consortium · BT · Airbus · ORCA
Post-Quantum	PQC solutions (pre-NIST, identity + comms focused)	Private	IETF standard · NATO/US DoD products · Lattice-based

Post-Quantum Cryptography & Security

Company	Technology	Stage	Key Highlight
PQShield ★	PQC hardware IP + embedded crypto (IoT/automotive)	Private (Series B)	NIST FIPS 203/204/205 implementations · 5KB RAM
Arqit Quantum ★	Quantum-safe symmetric encryption (SKA-Platform)	NASDAQ: ARQQ	NCSC PQC pilot · Encryption Intelligence Jan 2026
Crypto Quantique ★	QDID quantum tunneling for IoT device identity (QRNG)	Private (£10M+)	QuarkLink platform · IoT security at scale

Defense, Research & Sensing

Company	Technology	Stage	Key Highlight
QinetiQ ★	Defense quantum sensing, computing research (FTSE 250)	Public (QQ.L)	OQC military comms partner · DSTL quantum programme
Oxford Instruments ★	Quantum hardware infrastructure (Triton dilution fridges)	Public (OXIG.L)	Dilution fridges for IBM/Google/IonQ · NanoScience division
QLM Technology ★	Quantum gas lidar for methane/GHG emissions monitoring	Private (£12M A)	Bristol spinout · SLB lead · Project Canary US expansion

Government & Infrastructure

Company	Technology	Stage	Key Highlight
NQCC ★	National Quantum Computing Centre, Harwell — UK's sovereign QC hub	Gov't (UKRI)	Google Willow cloud access · IBM QS2 · Rigetti · OQC
UKRI / Innovate UK	Primary funder: National Quantum Tech Programme £1B+	Gov't Agency	Quantum Missions Pilot £15M+ · Quantum Data Centre of the Future
University of Oxford	Quantum spinout engine: OQC, Oxford Ionics,	University	Oxford Quantum Group · Beecroft · most spinouts

Company	Technology	Stage	Key Highlight
	PQShield, KETS		globally
University of Cambridge	Quantinuum Cambridge HQ · Nu Quantum · Riverlane	University	Cavendish Lab · Cambridge Quantum (now Quantinuum SW)
GCHQ / NCSC	UK national quantum security strategy · PQC migration mandate	Gov't Security	NCSC PQC pilot (Arqit) · £160M cybersecurity investment

★ = Detailed company profile included · 75+ active quantum technology organisations across the UK (March 2026)

Market Overview

£1B+ NQTP Investment since 2014	\$10B Quantinuum Valuation	\$1.075B Oxford Ionics Exit (IonQ)	£670M New Govt Commitment 2025
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UK National Quantum Strategy

The United Kingdom’s quantum strategy is built on three decades of academic quantum physics leadership and a National Quantum Technologies Programme (NQTP) that has operated continuously since 2014. The NQTP has invested over £1 billion through four Quantum Technology Hubs covering computing, communications, sensing, and enhanced timing, creating a research-to-commercialisation pipeline unmatched in Europe.

June 2025 Quantum Investment: The UK government committed £670 million over 10 years as part of a broader £1 billion+ frontier technology package—explicitly motivated by concern over foreign acquisitions of UK quantum assets (Oxford Ionics by IonQ, quantum unit of Oxford Instruments by US entities). This investment signals a strategic pivot from research funding to sovereign industrial capability.

National Quantum Computing Centre (NQCC): The NQCC at Harwell provides the UK’s sovereign quantum computing infrastructure, hosting systems from OQC, Rigetti, IBM Quantum System Two, and providing cloud access to Google’s Willow chip. It serves as the anchor institution for the UK’s ‘1 million quantum operations by 2028’ national target.

NCSC PQC Mandate: £160 million was invested in quantum technology to improve UK responses to crime, fraud, and money-laundering. The National Cyber Security Centre (NCSC) has launched a Post-Quantum Cryptography pilot programme (including Arqit) and issued mandatory guidance for government systems to transition to NIST PQC standards by 2030.

Oxford–Cambridge Spinout Engine: Oxford and Cambridge universities produce more quantum spinouts per year than any other institutions globally. Oxford Science Enterprises is the leading quantum investor by number of investments in the UK, with positions in OQC, Quantum Motion, PQShield, and multiple pre-Series A companies.

Competitive Landscape Summary

Company	Technology	Stage	Funding	Key Partners
Quantinuum	Trapped-ion	Private \$10B	\$625M+	Microsoft, NVIDIA, JPMorgan
OQC	Superconducting	Private \$100M+	\$100M+	AWS, NVIDIA, Digital Realty
ORCA	Photonic (room-temp)	Private	\$25M	BT, Airbus, BP, Innovate UK
Universal Q	Trapped-ion micro.	Private	£100M+	Sussex, NQCC, EU HPC
Quantum Motion	Silicon CMOS	Private	£62M+	Bosch Ventures, DARPA
Riverlane	QEC middleware	Private \$75M SC	\$75M+	Rigetti, IQM, NQCC, Altair
Nu Quantum	QC Networking hw	Private \$60M SA	\$60M+	National Grid, Cambridge
PQShield	PQC hardware IP	Private S-B	£30M+	Arm, TSMC, Samsung Next
Arqit	Quantum-safe crypto	NASDAQ: ARQQ	\$37M cash	NATO, NCSC, RAD, Sparkle
QinetiQ	Defense quantum	FTSE 250	£1.9B rev	UK MoD, DSTL, Five Eyes
Oxford Inst.	QC infrastructure	FTSE All-Share	£600M+ rev	IBM, Google, IonQ, all
QLM Tech	Quantum gas sensing	Private £12M	SLB-led	Project Canary, Innovate UK

Company Profiles

1. Quantinuum

Company	Quantinuum Ltd (formed 2021: Cambridge Quantum + Honeywell Quantum Solutions)
Headquarters	Cambridge, UK + Broomfield, Colorado, USA
CEO	Rajeeb Hazra (succeeded Ilyas Khan, who remains Vice Chairman)
Technology	Trapped-ion QCCD (Quantum Charge-Coupled Device) H-Series hardware + full-stack software
Helios System	Single-qubit gate fidelity 99.9975% across 98 physical qubits (94 logical) — world record
Valuation	\$10 billion (private) — confidential S-1 filed January 2026 for potential IPO at \$20B+ target
Key Funding	\$625M+ total (incl. \$600M Sep 2025: NVIDIA, JPMorgan, Quanta Computer, Fidelity, Mitsui)
Key Partners	Microsoft (800x error reduction), NVIDIA (NVQLink), JPMorgan, SoftBank, STFC Hartree
DARPA	QBI Stage B — Apollo fault-tolerant roadmap by 2029

Technology

Quantinuum operates the highest-fidelity quantum computer commercially available. Its Helios H-Series system achieved a single-qubit gate fidelity of 99.9975% across 98 physical qubits (94 logical) — a world record that establishes Quantinuum in a distinct performance tier from all competitors. The QCCD architecture shuttles ions between interaction zones using precisely controlled electric fields, enabling all-to-all qubit connectivity that superconducting systems cannot easily replicate. Quantum volume exceeding 1,048,576 — orders of magnitude above any competitor — means Quantinuum's systems can reliably execute deeper circuits on more qubits.

Quantinuum's software stack extends far beyond hardware: InQuanto (quantum chemistry), TKET (compiler/SDK), Lambeq (quantum natural language processing), and Quantum Origin (quantum-enhanced QRNG for cryptographic seeding) each address distinct commercial verticals. The company's 2025 partnership with Microsoft produced an 800x error rate reduction using Microsoft's QEC applied to Quantinuum hardware — one of the most significant quantum error correction results published.

Recent Milestones (2024 — March 2026)

- Helios system: 99.9975% single-qubit gate fidelity, 98 physical qubits (94 logical) — world record for trapped-ion systems
- \$600M funding round (September 2025): NVIDIA NVentures, JPMorgan Chase, Quanta Computer, Fidelity, Mitsui, SoftBank — \$10B pre-money valuation
- Confidential S-1 filed (January 2026) for potential IPO — target valuation reported at \$20B+, would be the largest quantum IPO in history

- DARPA QBI Stage B advancement — Quantinuum competing alongside IonQ, PsiQuantum, QuEra for utility-scale validation
- Microsoft 800x error reduction: logical error rates reduced 800-fold using Microsoft QEC on Quantinuum hardware
- Qatar joint venture (\$1B): partnering with Qatar Investment Authority for Middle East quantum infrastructure
- Singapore Helios installation planned for 2026 — first Quantinuum system in Asia
- Apollo roadmap: universal fault-tolerant quantum computer targeting 2029 — aligned with IBM's Starling and Google's 2029 target

Strategic Position for Korea

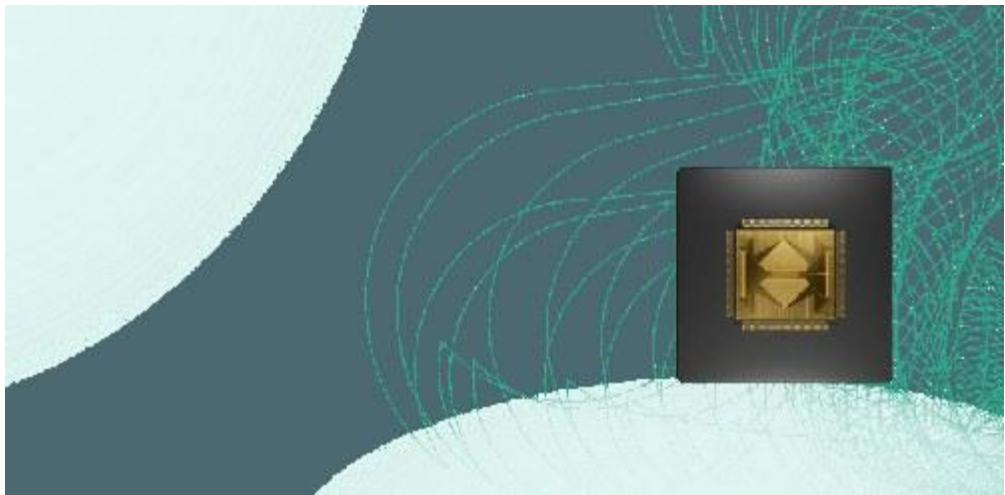
Quantinuum's world-record fidelity makes it the preferred hardware for quantum chemistry, drug discovery, and materials science — all priority applications for Korean conglomerates. Samsung SDI and LG Energy Solution (battery materials), Hanwha (aerospace materials), and Yuhan/Boryung (pharmaceutical) are natural Quantinuum enterprise customers. The planned Singapore deployment signals that Quantinuum is actively building APAC infrastructure, and Korea would be a logical next step given KAIST's quantum chemistry research alignment with InQuanto.

System Model H2

Challenging the limits of
classical computing



Quantinuum H2 — micro-fabricated gold ion trap chip (QCCD architecture, world record quantum volume > 1,048,576)



Quantinuum H2 full system — trapped-ion quantum computing system, highest-fidelity QC commercially available (2025)

2. Oxford Quantum Circuits (OQC)

Company	Oxford Quantum Circuits Ltd (OQC)
Founded	2017 — Oxford, UK (University of Oxford spinout, Dr Peter Leek)
CEO	Ilana Wisby
Technology	Superconducting qubits — Coaxmon 3D architecture (Dimon, Trimon generations), Toshiko system
Funding	\$100M+ raised (Series A £38M 2022 + Series B led by SBI Investment 2023)
2025 Deploy	First European deployment in US data centre: Digital Realty New York (September 2025), integrated with NVIDIA
2026 Roadmap	Genesis system: 16 logical qubits, 1,000 operations/sec, hardware-level error detection — commercial 2026
Long-term	200 logical qubits by 2028 · 50,000 logical qubits by 2034 (most ambitious public quantum roadmap)

Technology

OQC’s Coaxmon architecture places qubit components on opposite sides of a substrate, dramatically simplifying wiring and improving coherence compared to traditional planar superconducting designs. The Toshiko system — OQC’s current commercial platform — is deployed in standard commercial data centres without requiring dedicated quantum facilities, making it the most accessible enterprise-grade superconducting system available. In February 2026, OQC published a preprint demonstrating wafer-scale packaging supporting over 500 superconducting qubits on a single 3-inch die with a median T_1 of 97 μs — a manufacturing breakthrough that validates OQC’s path to its 2028 roadmap.

The Genesis system (planned 2026) introduces OQC’s patented dual-rail Dimon qubit design,

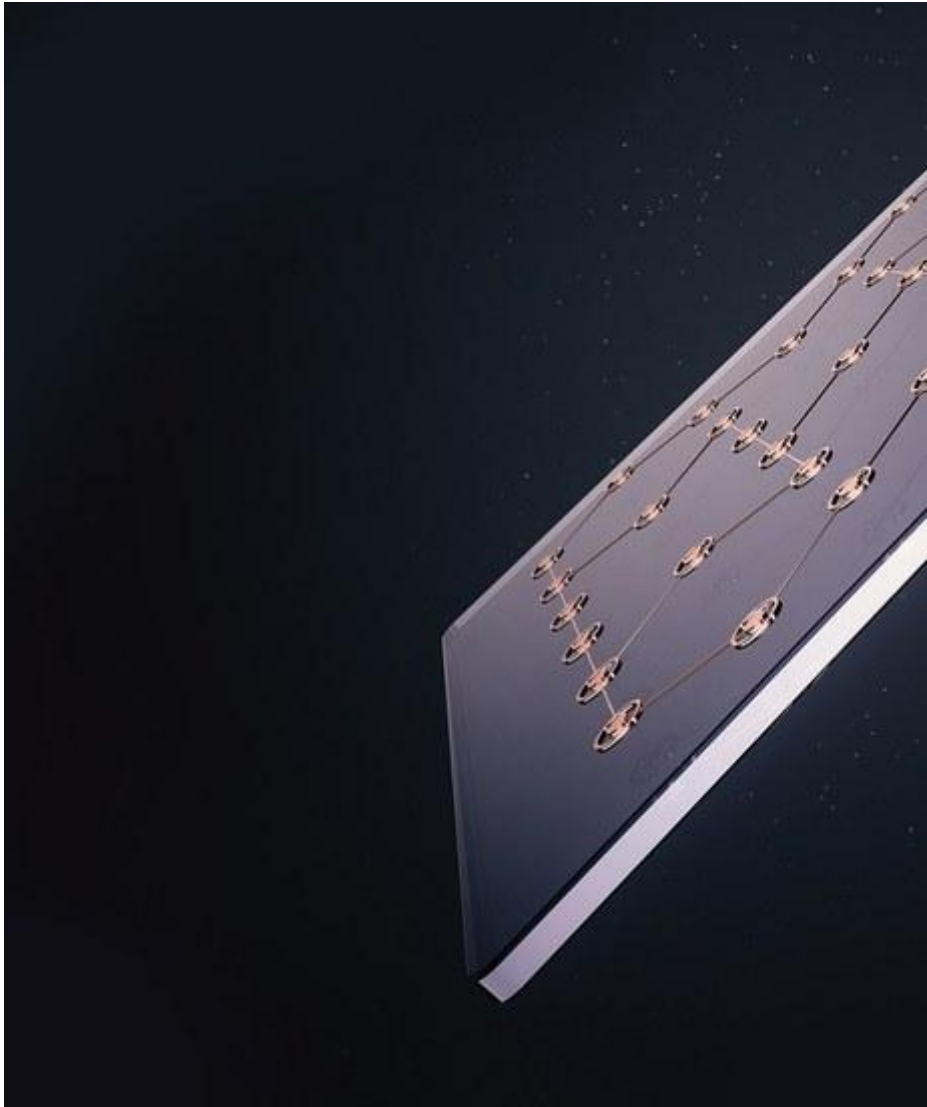
which encodes a logical qubit in the footprint of a single physical qubit — a dramatic efficiency improvement. Genesis targets “KiloQuOp” scale: 1,000 quantum operations per second with hardware-level error detection. This would make it the first commercially available quantum computer with built-in error detection at the qubit level.

Recent Milestones (2024 — March 2026)

- Digital Realty New York deployment (September 2025): first European quantum system in US commercial data centre; integrated with NVIDIA GPU infrastructure for hybrid quantum-classical workflows
- 500+ qubit wafer-scale packaging demonstrated (February 2026): single 3-inch die, median T_1 97 μ s — validates path to 200 logical qubits by 2028
- Genesis system development: 16 logical qubits, dual-rail Dimon qubits, hardware-level error detection — commercial release targeted 2026
- OQC-Riverlane QEC integration (2025): first real-time QEC decoder integration in UK, combining OQC hardware with Riverlane’s Deltaflow middleware
- QinetiQ partnership: quantum analysis of military communications networks — UK defence customer engagement
- Full 9-year public roadmap (2025-2034): most detailed publicly disclosed quantum roadmap, targeting 50,000 logical qubits by 2034

Strategic Position for Korea

OQC’s data-centre-native deployment model is uniquely relevant for Korea. Korean hyperscale data centres (Kakao, Naver, Samsung SDS, KT) operate in standard rack environments where OQC’s room-temperature-compatible systems can deploy without infrastructure changes. The NVIDIA GPU integration positions OQC for hybrid quantum-HPC workloads at Korean HPC operators KISTI and KREONET.



OQC Coaxmon chip architecture — 3D superconducting qubit design with components on opposite sides of substrate, enabling data-centre deployment without cryogenic facility (Oxford 2025)

3. ORCA Computing

Company	ORCA Computing Ltd
Founded	2019 — London, UK (Oxford University spinout)
CEO	Richard Murray
Technology	Photonic quantum computing — telecom-standard single photons + proprietary quantum memories
Key Feature	Room-temperature operation: no cryogenic cooling required for most components; rack-mountable
Funding	\$25.1M total (Series A \$15M + Innovate UK grants)

Consortium	Leads 14-organisation 'Quantum Data Centre of the Future': BT, BP, Airbus, KETS, Riverlane, PQShield
2026 Product	PT-3: commercial-release hybrid quantum-classical accelerator targeting ML/AI workloads

Technology

ORCA's photonic approach uses telecom-grade single photons stored in proprietary quantum memories — a combination that enables rack-mountable, air-cooled quantum computers that can be deployed in standard data centre environments alongside classical hardware. Unlike superconducting or ion trap systems requiring millikelvin cooling, ORCA's systems operate near room temperature, dramatically reducing deployment complexity and cost.

The PT-series product line—PT-1 (research), PT-2 (mid-scale), PT-3 (commercial, 2026)—targets hybrid quantum-classical workloads for machine learning, optimization, and simulation. Integration with Python, PyTorch, and NVIDIA's CUDA-Q toolchain means ORCA's systems slot into existing ML infrastructure rather than requiring new quantum-specific tooling. The Quantum Data Centre of the Future consortium, backed by Innovate UK, positions ORCA at the centre of the UK's quantum-classical HPC integration strategy.

Recent Milestones (2024 — March 2026)

- PT-3 commercial launch targeted 2026: positioning as hybrid quantum-classical accelerator for ML workloads in data centres
- Quantum Data Centre of the Future: leading 14-organisation Innovate UK consortium (£11.6M) with BT, BP, Airbus, KETS Quantum, Riverlane, PQShield, and 10 UK universities
- QUICHE project (Innovate UK): alongside Riverlane and Quantum Motion, targeting quantum chemistry workflows for materials science
- NVIDIA CUDA-Q integration: ORCA systems compatible with NVIDIA's quantum-classical hybrid computing framework
- PyTorch/Python native: quantum algorithms deployable using familiar ML toolchains — lowest quantum adoption barrier of any hardware company

Strategic Position for Korea

ORCA's room-temperature, rack-mountable systems are the most deployable quantum hardware for Korean enterprises. Korean data centre operators can integrate PT-3 systems without infrastructure changes, enabling immediate quantum experimentation. The ML/AI workload focus aligns with Korean AI infrastructure priorities at Kakao, Naver, and Samsung SDS.



ORCA Computing PT-2 — photonic quantum computer in data centre rack environment, room-temperature operation without cryogenic cooling (London 2025)

4. Universal Quantum

Company	Universal Quantum Ltd
Founded	2018 — Brighton, UK (University of Sussex spinout)
Co-founders	Prof. Winfried Hensinger + Dr Sebastian Shermer
Technology	Trapped-ion quantum computing using microwave technology — no lasers required
Innovation	Modular chip architecture: quantum modules connected by electric field links for scalability
Funding	£100M+ raised including £75M+ from industrial and strategic investors
Key Partners	German Aerospace Center (DLR), UK NQCC, European HPC infrastructure
Distinction	First company to demonstrate scalable trapped-ion system using microwave (not laser) control

Technology

Universal Quantum’s microwave approach to trapped-ion quantum computing eliminates the most complex and fragile component of conventional trapped-ion systems: the precision laser setup. Traditional trapped-ion systems (IonQ, Quantinuum) require a laboratory full of precisely aligned laser beams and optical components. Universal Quantum replaces all of this with microwave radiation — the same technology used in mobile phones — applied through chip-integrated antennas. This makes their systems dramatically simpler to manufacture, operate, and scale.

The modular architecture links multiple quantum processor chips using electric field links —

allowing independent modules to be produced and then connected, rather than requiring a single monolithic chip to scale. This approach mirrors successful classical computing architectures (AMD’s chiplets, Intel’s EMIB interconnects) and could enable fault-tolerant quantum systems by linking smaller high-quality modules rather than scaling a single large imperfect chip.

Recent Milestones (2024 — March 2026)

- Full microwave ion trap system demonstrated: first-of-kind trapped-ion quantum computer operating entirely without lasers
- DLR (German Aerospace Center) partnership: joint quantum computing programme for aerospace simulation and optimization
- EU HPC integration: Universal Quantum systems planned for integration with European HPC infrastructure through EuroHPC
- NQCC deployment: system installed at UK National Quantum Computing Centre, Harwell
- Scaling roadmap: targeting 100+ qubit systems through modular microwave chip architecture in 2026-2027

5. Quantum Motion

Company	Quantum Motion Technologies Ltd
Founded	2017 — London, UK
CEO	John Morton
Technology	Silicon spin qubits using CMOS-compatible manufacturing processes
Key Advantage	Qubits fabricated using existing semiconductor foundry infrastructure — same fabs as smartphone chips
Funding	£62M+ including £42M led by Bosch Ventures
DARPA	QBI programme participant — silicon qubit approach validated for utility-scale research
Partners	Bosch Ventures (strategic), UCL, EPSRC (SiQEC project — silicon QEC demonstration)

Technology & Strategic Position

Quantum Motion’s silicon spin qubit approach is the most manufacturable path to quantum computing — by building qubits in CMOS (Complementary Metal-Oxide-Semiconductor) processes already used to manufacture billions of transistors, Quantum Motion can leverage the entire classical semiconductor supply chain. UK foundry partnerships (IMEC, UK semiconductor manufacturers) and the Bosch Ventures investment signal that automotive and industrial giants view silicon spin qubits as the most realistic path to quantum hardware at scale.

The SiQEC project (Innovate UK Quantum Missions) targets error correction demonstration in spin-based quantum computing alongside UCL — a key milestone for demonstrating that silicon qubits can support the QEC required for fault-tolerant operation. DARPA QBI participation validates the approach at the highest level of government quantum scrutiny.

- Bosch Ventures £42M lead investment: automotive giant betting on silicon spin qubits for industrial quantum applications
- DARPA QBI participant: silicon qubit approach validated alongside Quantinuum, IonQ, IBM, and PsiQuantum
- SiQEC (Innovate UK): demonstrating quantum error correction in silicon spin qubits with UCL
- CMOS manufacturing advantage: Samsung Foundry, TSMC, GlobalFoundries could all manufacture Quantum Motion's qubit chips using existing processes
- QUICHE project: alongside ORCA and Riverlane, targeting quantum chemistry applications for materials science

6. Oxford Ionics (Acquired by IonQ, June 2025)

Company	Oxford Ionics Ltd (Acquired by IonQ Inc., June 2025)
Founded	2019 — Oxford, UK (University of Oxford spinout)
Technology	Ion-trap-on-a-chip: control electronics integrated directly on chip (no external lasers/microwaves)
Exit	\$1.075 billion (IonQ acquisition, June 2025) — largest UK quantum exit in history
DARPA	QBI Stage A — selected alongside Quantinuum as highest-priority trapped-ion systems
Legacy	IonQ acquired Oxford Ionics' chip-scale ion trap technology as path to 2M+ physical qubits by 2030
Significance	Proved UK quantum spinout-to-acquisition pipeline — catalysed £670M UK govt investment decision

Technology & Historical Significance

Oxford Ionics developed ion-trap-on-a-chip technology that placed all qubit control electronics directly on the chip itself, eliminating the external laser or microwave infrastructure that conventional trapped-ion systems require. This chip-scale integration was the key to scaling trapped-ion systems beyond laboratory configurations — IonQ cited Oxford Ionics' technology as the basis for its roadmap to over two million physical qubits by 2030.

The \$1.075 billion acquisition (June 2025) was a watershed moment for the UK quantum ecosystem. It proved that British quantum spinouts can achieve unicorn exits through technology acquisition at the highest level of the global quantum market, catalysed the UK government's subsequent £670 million quantum investment commitment, and demonstrated that DARPA validation (Oxford Ionics was a QBI Stage A participant) creates acquisition interest from US quantum leaders.

- \$1.075B acquisition by IonQ (June 2025): largest UK quantum exit; largest in global quantum industry 2025
- DARPA QBI Stage A: one of only two trapped-ion companies selected (alongside Quantinuum) for highest-priority validation

- 2M qubit roadmap contribution: Oxford Ionics’ chip-scale ion trap is the technology behind IonQ’s path to 2 million physical qubits by 2030
- UK ecosystem catalyst: acquisition triggered £670M government quantum investment and reinforced UK spinout-to-acquisition model
- Team now operating as IonQ UK R&D Hub in Oxford — creating UK-based quantum hardware research anchor

7. Riverlane

Company	Riverlane Ltd
Founded	2016 — Cambridge, UK (University of Cambridge spinout, Dr Steve Brierley)
CEO	Steve Brierley
Technology	Quantum error correction middleware — Deltaflow QEC stack (hardware-agnostic)
Key Product	Deltaflow: QEC decoder chip + software processing error syndromes in real time, billions/second
Funding	\$75M Series C (Planet First Partners, Cambridge Innovation Capital, NSSIF, Altair)
Reach	Partners with 60%+ of all quantum computer companies globally — Rigetti, IQM, NQCC and many more
Report	QEC Report 2025: 120 peer-reviewed QEC papers in 10 months (vs 36 in 2024) — global talent warning

Technology

Riverlane occupies the quantum computing stack’s most critical and underappreciated layer: error correction. Every path to fault-tolerant quantum computing requires QEC — a continuous process of detecting and correcting errors faster than they accumulate. Riverlane’s Deltaflow stack combines dedicated QEC decoder chips (proprietary ASICs), real-time error syndrome processing software, and a hardware-agnostic middleware layer that works across all major qubit modalities: superconducting (IBM, Rigetti, OQC), trapped-ion (Quantinuum, IonQ), neutral atom (PASQAL, QuEra), and silicon spin (Quantum Motion).

This hardware-agnostic positioning is Riverlane’s structural moat. Unlike hardware companies that bet on a single qubit modality, Riverlane profits from quantum hardware development regardless of which approach wins the hardware race — every quantum computer will need QEC, and Deltaflow is designed to be the universal QEC layer. The 60%+ partnership rate across quantum hardware companies confirms this positioning is working.

Recent Milestones (2024 — March 2026)

- \$75M Series C (2024): Planet First Partners lead, Cambridge Innovation Capital, NSSIF (National Security Strategic Investment Fund), Altair — expands QEC team to 100+ specialists
- QEC Report 2025 (November 2025): found 120 peer-reviewed QEC papers in first 10 months of 2025 (vs 36 in 2024) — identified global talent shortage of up to 16,000 QEC specialists needed by 2030

- Rigetti real-time QEC breakthrough (October 2024): first demonstration of low-latency real-time QEC on Rigetti hardware using Deltaflow decoder
- OQC integration (2025): first real-time QEC decoder integration in the UK — combined OQC Toshiko hardware with Deltaflow middleware
- NQCC national integration: Deltaflow stack integrated into UK National Quantum Computing Centre Harwell infrastructure
- Quantum Data Centre of the Future consortium: Riverlane contributing QEC layer to 14-organisation BT/ORCA-led UK quantum data centre project

Strategic Position for Korea

Riverlane’s hardware-agnostic QEC is the highest-leverage entry point into Korea’s quantum ecosystem. Rather than betting on a single quantum hardware vendor, Korea can deploy Riverlane’s Deltaflow stack across whichever hardware platform is adopted — IBM Quantum, IonQ, Quantinuum, or PASQAL. KISTI and KAIST represent natural Riverlane partners for Korean QEC research and deployment.

8. Nu Quantum

Company	Nu Quantum Ltd
Founded	2018 — Cambridge, UK (University of Cambridge Cavendish Laboratory spinout)
CEO	Carmen Palacios-Berraquero
Technology	Quantum networking hardware: entanglement distribution, optical switching, quantum networking units
Key Milestone	\$60M Series A (December 2025): led by National Grid — largest early-stage quantum funding in UK history
Lab	February 2026: opened Europe’s first industrial trapped-ion quantum networking laboratory in Cambridge
Mission	Enable modular quantum computing: connect smaller processors into large-scale distributed systems
Partners	National Grid (strategic), Cisco (HyperIon project), University of Sussex

Technology & Strategic Position

Nu Quantum builds the networking hardware layer that modular quantum computing requires. As quantum processors scale toward millions of qubits, single-chip architectures face fundamental physical limits — the solution is connecting multiple smaller processors over quantum networks. Nu Quantum’s networking units handle the entanglement distribution, optical switching, and quantum state transfer required to link quantum processors at scale.

National Grid’s lead investment in Nu Quantum’s \$60M Series A is strategically significant: it signals that major infrastructure operators see quantum networking as critical future infrastructure for energy grid management, secure communications, and distributed computing. The February 2026 opening of Europe’s first industrial trapped-ion quantum networking

laboratory in Cambridge establishes Nu Quantum as the anchor institution for quantum networking R&D across Europe.

- \$60M Series A (December 2025): National Grid lead investor — largest early-stage quantum raise in UK history; supports product development and EU/US expansion
- Europe's first industrial quantum networking lab (February 2026): Cambridge facility dedicated to trapped-ion-based quantum networking hardware R&D
- Hyperlon project (Innovate UK): with Cisco and University of Sussex, developing qubit-photon interface for distributed quantum computing
- National Grid strategic investor: quantum networking as future energy infrastructure — distributed quantum computing for grid optimization
- Quantum internet vision: Nu Quantum's networking stack targets the scaling challenge facing all quantum hardware — modular architecture via networked processors



Nu Quantum — photonic quantum networking hardware in Cambridge laboratory (entanglement distribution system for modular quantum computing, 2025)

9. PQShield

Company	PQShield Ltd
Founded	2018 — Oxford, UK (University of Oxford spinout)
CEO	Ali El Kaafarani
Technology	Post-quantum cryptography (PQC) hardware IP and embedded software for silicon chips
Key Products	Hardware IP implementing NIST FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA) on silicon
Funding	£30M+ (Series B) — investors include Arm, Addition, Oxford Science

	Enterprises, Elk VC
Market	IoT device security, automotive ECUs, defence hardware, 5G baseband chips
Distinction	5KB RAM PQC implementation (Embedded World 2026) — world's smallest verified PQC hardware footprint

Technology & Strategic Position

PQShield's core product is hardware IP — semiconductor building blocks that chip designers embed into their silicon to add PQC capabilities. Rather than building complete chips, PQShield licenses its IP to semiconductor companies (Arm, TSMC supply chain partners) who then integrate it into their own chips. This IP model has enormous leverage: once embedded, PQShield's PQC is in every device that uses the chip, from IoT sensors to automotive ECUs to defence hardware.

The August 2024 NIST PQC standards (FIPS 203/204/205) triggered a mandatory transition that creates a multi-billion-dollar market for PQShield's technology. Every government, enterprise, and connected device system must migrate to these new standards — PQShield's hardware IP is the fastest path to compliance for chip manufacturers. Arm's investment is particularly strategic: Arm architecture underlies nearly every smartphone and IoT device, meaning PQShield's IP could reach billions of deployed devices.

- Series B raised: investors include Arm (strategic — signalling PQShield IP planned for Arm-based chips), Addition, Oxford Science Enterprises
- NIST FIPS 203/204/205 implementations: PQShield is one of the few companies with verified silicon implementations of all three final NIST PQC standards
- 5KB RAM PQC (Embedded World 2026): world's smallest verified PQC hardware footprint — enables PQC on the most constrained IoT devices
- Quantum Data Centre consortium: PQShield contributing PQC layer to ORCA's Quantum Data Centre of the Future project
- NCSC endorsement: UK National Cyber Security Centre certified PQShield as part of Post-Quantum Cryptography pilot



PQShield — post-quantum cryptography semiconductor chip implementing NIST FIPS 203/204/205 standards for IoT and automotive applications (2025)

10. Arqit Quantum

Company	Arqit Quantum Inc.
Founded	2017 — London, UK (NASDAQ: ARQQ since 2021)
CEO	Andy Leaver
Technology	Quantum-safe symmetric key agreement (SKA-Platform) — software-based, no quantum hardware needed
Revenue	FY2025: \$530K (year ended Sep 2025) — transitioning to SaaS licensing model
New Product	Encryption Intelligence (Jan 2026): cryptographic inventory + PQC migration planning product
Key Customers	Government agencies (EMEA multi-year), Sparkle (optical transport layer POC Athens)
NCSC	Selected for UK National Cyber Security Centre Post-Quantum Cryptography Pilot

Technology & Strategic Position

Arqit's SKA-Platform generates quantum-safe symmetric encryption keys without relying on the mathematical assumptions (factoring difficulty, discrete logarithm) that quantum computers could break. Instead of quantum hardware, the platform uses software agents deployed on endpoints that generate cryptographically secure keys locally through a quantum-safe protocol. This approach is deployable today on existing infrastructure — no quantum hardware required at either end.

The January 2026 launch of Encryption Intelligence represents Arqit's strategic pivot from hardware-dependent quantum security to enterprise PQC migration advisory. Encryption Intelligence provides automated cryptographic discovery (identifying all encryption in use across a network), continuous monitoring, and risk prioritisation for PQC migration planning — the first commercial product to automate the 'cryptographic inventory' step that every enterprise must complete before migrating to NIST PQC standards.

- Encryption Intelligence commercial launch (January 2026): automated cryptographic discovery and PQC migration planning — directly addresses the NIST PQC transition mandate
- NCSC PQC Pilot selected: UK government cybersecurity agency validation of Arqit's discovery and migration planning methodology
- Sparkle POC (December 2025): quantum-resistant encryption on optical transport layer across metropolitan ring in Athens — L1 encryption with no hardware changes
- RAD collaboration (March 2026): joint quantum-safe encryption for telcos — combining Arqit's key generation with RAD's carrier edge platform
- \$36.9M cash (September 2025): sufficient runway for SaaS transition; FY2026 contracted revenue ~\$1.2M

11. KETS Quantum Security

Company	KETS Quantum Security Ltd
Founded	2017 — Bristol, UK (University of Bristol spinout)
CEO	Chris Erven
Technology	Integrated photonics QKD hardware — quantum key distribution on photonic integrated circuits
Key Feature	Chip-scale QKD: miniaturises QKD into a photonic IC the size of a fingernail
Consortium	Quantum Data Centre of the Future: led architecture/communications alongside ORCA, BT, Airbus, PQShield
Demonstration	Hybrid QKD/PQC link + entropy-as-a-service with PQShield in UK quantum data centre project
Market	Fibre-optic network operators, government secure comms, critical national infrastructure

Technology & Strategic Position

KETS' integrated photonics approach miniaturises the components of a quantum key distribution system onto a single photonic chip — replacing a table full of optical components with a fingernail-sized device. This integration dramatically reduces the cost, size, and

complexity of QKD deployment, enabling it to be incorporated into standard network equipment rather than requiring dedicated quantum communication infrastructure.

The hybrid QKD/PQC demonstration (within the Quantum Data Centre of the Future consortium) is particularly significant: it shows that QKD (for the highest security scenarios) and PQC (for scalable, software-deployable security) can work together — providing defence-in-depth against quantum attacks. For Korean telcos (KT, SKT, LG U+) and financial infrastructure operators, this hybrid approach addresses both near-term PQC migration requirements and long-term quantum-proof security needs.

- Quantum Data Centre of the Future: led architecture and communications workstream — first demonstration of integrated QKD in a quantum data centre context
- Hybrid QKD/PQC link demonstrated: combining quantum key distribution with post-quantum cryptography for layered quantum-safe security
- Entropy-as-a-service with PQShield: quantum random number generation as a cloud service feeding PQC key generation
- Innovate UK awards: UP-SENSE project developing next-generation quantum gas sensors using photon up-conversion and SPAD detectors



KETS Quantum Security — OKD-TX QKD circuit board (integrated photonics quantum key distribution hardware, Bristol 2025)

12. Crypto Quantique

Company	Crypto Quantique Ltd
Founded	2017 — London, UK
CEO	Shahram Mossayebi
Technology	QDID: quantum tunnelling in semiconductor devices generates unclonable hardware device identities

Key Product	QuarkLink: IoT device lifecycle management (onboarding, certificates, key management) at scale
Market	IoT security, semiconductor IP, connected device identity management
Distinction	Uses fundamental quantum behaviour (electron tunnelling) to create truly unique device identities without external key injection
Partners	IoT platform vendors, semiconductor IP providers, global IoT security ecosystem

Technology & Strategic Position

Crypto Quantique's QDID (Quantum-Driven Identifiers) technology exploits quantum mechanical electron tunnelling in standard semiconductor devices to generate device identities that are physically unclonable — no two devices produce the same identity even when manufactured from identical designs. This solves a fundamental IoT security problem: how to provision billions of devices with unique, secure identities without a key injection step that creates a supply chain vulnerability.

The QuarkLink platform manages the full lifecycle of these quantum-derived device identities at scale — enrollment, certificate management, key rotation, and decommissioning. For Korea, where Samsung and LG manufacture billions of IoT-connected devices annually, QDID+QuarkLink provides a security layer that can be embedded at the chip design stage and managed throughout product lifetime.

- QDID technology: quantum tunnelling generates physically unclonable device fingerprints in standard CMOS processes — no dedicated quantum hardware
- QuarkLink platform: managing millions of IoT device identities through certificate lifecycle and key management
- Samsung supply chain relevance: QDID compatible with Samsung Semiconductor processes — direct applicability to Korea's IoT device export sector
- Series A raised: investors include Samsung Catalyst Fund (strategic — direct Korea connection)

13. Phasecraft

Company	Phasecraft Ltd
Founded	2019 — London, UK (UCL + University of Bristol spinout)
CEO	Ashley Montanaro
Technology	Quantum algorithms for near-term hardware — materials science, pharmaceutical R&D, energy
Key Partners	Google (quantum hardware access), BASF (materials), AstraZeneca (pharma), Bloomberg (finance)
Focus	Making NISQ-era quantum computers commercially useful NOW through algorithm optimisation
Funding	£10M+ from IP Group, Parkwalk, UCL Technology Fund

Distinction

Published the tightest known bounds on quantum chemistry algorithms — most rigorous quantum algorithm research in UK

Technology & Strategic Position

Phasecraft occupies the most commercially pragmatic position in the UK quantum software ecosystem: making existing NISQ-era quantum computers deliver real value before fault tolerance arrives. Its algorithms team — led by some of Europe's top quantum computing theorists — has developed techniques that reduce the circuit depth required to simulate materials and chemicals by up to 100x compared to naive implementations, enabling commercially useful quantum chemistry on today's imperfect hardware.

The Google hardware partnership provides Phasecraft access to Sycamore and Willow systems, validating their algorithms on the most advanced quantum hardware available. BASF and AstraZeneca partnerships represent production-grade enterprise engagements where quantum-enhanced simulations are being used to inform real materials and pharmaceutical research decisions.

- Google hardware partnership: Phasecraft algorithms validated on Google Sycamore and Willow quantum processors
- BASF materials research: quantum-enhanced simulation of battery and catalysis materials chemistry
- AstraZeneca pharmaceutical R&D: quantum algorithms for drug-protein binding simulation
- Bloomberg quantum finance: quantum algorithms for portfolio optimization and risk analysis
- Published tightest bounds on quantum chemistry resource estimates: most rigorous near-term quantum algorithm research in Europe

14. QuantrolOx

Company	QuantrolOx Ltd
Founded	2022 — Oxford, UK (University of Oxford spinout)
CEO	Natalia Ares
Technology	Machine learning-based automated quantum control — AI tunes and operates quantum devices
Key Product	Quantum device automation: ML replaces manual calibration and operation of quantum processors
Partners	Intel, national quantum labs, all major qubit platform vendors
Distinction	First company to fully automate quantum dot calibration using deep reinforcement learning
India JV	Q-HUB@MAHE — joint venture with Manipal Academy of Higher Education (March 2026)

Technology & Strategic Position

QuantrolOx addresses one of quantum computing’s most overlooked practical challenges: quantum devices are extraordinarily difficult to calibrate and operate. Even after building a quantum processor, operators must manually tune hundreds of parameters — a process that takes hours and must be repeated frequently as conditions drift. QuantrolOx’s machine learning algorithms automate this calibration process, reducing human involvement from hours to minutes and enabling quantum systems to operate reliably at scale.

This control layer is hardware-agnostic — QuantrolOx’s software works with superconducting, spin, and other qubit types. Intel partnership validates the technology at the chip scale. The Q-HUB@MAHE joint venture (March 2026) with India’s Manipal Academy signals QuantrolOx’s APAC expansion strategy — Korea’s KAIST and POSTECH are natural next partners.

- Intel partnership: QuantrolOx ML algorithms deployed for automated calibration of Intel quantum dot devices
- Deep reinforcement learning for QD calibration: first published demonstration of fully autonomous quantum device operation
- Q-HUB@MAHE (March 2026): joint venture establishing quantum hub in India — APAC expansion model applicable to Korea
- Hardware-agnostic control: works with superconducting, spin qubit, and other platforms — universal quantum control layer

15. QinetiQ

Company	QinetiQ Group plc
Founded	2001 (spinout from UK Defence Evaluation and Research Agency, DERA)
Headquarters	Farnborough, Hampshire, UK
Revenue	£1.9 billion (FY2025, FTSE 250)
Technology	Defense quantum research: quantum sensing, quantum computing analysis, PQC for defense
Key Activity	Quantum analysis of military communications networks (with OQC) · DSTL quantum research partner
Defense Role	Technical advisor to UK MOD, Five Eyes alliance countries, and NATO on quantum technology
Distinction	Only UK publicly-traded pure defense company with active quantum technology programme

Technology & Strategic Position

QinetiQ occupies the UK’s most strategically important quantum defense position. As a FTSE 250 defense technology company spun out of the UK government’s Defense Evaluation and Research Agency, QinetiQ has embedded access to UK MOD quantum programs, Five Eyes intelligence partnerships, and classified quantum technology research. Its partnership with OQC on quantum analysis of military communications networks is the first public demonstration of enterprise quantum computing applied to UK defense use cases.

For Korea, QinetiQ’s defense quantum relevance is direct: UK-Korea defense cooperation

under the UK-ROK Global Strategic Partnership provides the framework for technology transfer in quantum sensing, quantum communications, and PQC for military systems. QinetiQ's expertise in GPS-denied navigation sensing (analogous to North Korea GPS jamming threat) and electromagnetic spectrum monitoring are directly applicable to ROK defense requirements.

- OQC partnership: quantum analysis of UK military communications networks — first defense quantum computing use case in UK
- DSTL (Defense Science and Technology Laboratory) quantum research partner: advising UK MOD on quantum technology acquisition strategy
- Five Eyes quantum engagement: QinetiQ advises US, Australia, Canada, and New Zealand on quantum technology defence applications
- PQC for defense: QinetiQ leading UK MOD transition to post-quantum cryptography across classified systems
- Quantum sensing: GPS-denied navigation and electromagnetic spectrum monitoring R&D directly applicable to ROK requirements

16. Oxford Instruments

Company	Oxford Instruments plc
Founded	1959 — Oxford, UK (FTSE All-Share)
Revenue	£600M+ (FY2025)
Technology	Quantum hardware infrastructure: Triton dilution refrigerators, cryogenic systems, NanoScience tools
Key Products	Triton dilution refrigerator: cools quantum chips to 10 millikelvin — used by IBM, Google, IonQ, Quantinuum
Market Share	~40% of all dilution refrigerators shipped to quantum computing customers globally
AUKUS	Triton systems deployed at quantum computing facilities across UK, US, Australia
Distinction	The most critical enabling infrastructure company in global quantum computing — without Triton, most QC doesn't work

Technology & Strategic Position

Oxford Instruments is the most underappreciated strategic company in the global quantum computing ecosystem. Its Triton dilution refrigerator — a device that cools superconducting quantum chips to 10 millikelvin (0.01 degrees above absolute zero) — is the essential enabling infrastructure for IBM, Google, IonQ (superconducting mode), Quantinuum, OQC, Rigetti, and virtually every other superconducting quantum computer company. Without Triton, quantum computing largely doesn't work.

With approximately 40% global market share in dilution refrigerators shipped to quantum computing customers, Oxford Instruments sits at the supply chain bottleneck of quantum computing scaling. Every quantum computer that ships increases demand for Triton systems — making Oxford Instruments one of the best quantum computing investments regardless of which

qubit modality wins the hardware race. The company's NanoScience division additionally provides scanning probe microscopes, vibration isolation systems, and other tools used in quantum chip characterisation and fabrication.

- Triton dilution refrigerator: ~40% global market share — IBM, Google, IonQ, Quantinuum, OQC, Rigetti all use Triton systems
- Quantum computing scaling lever: every new quantum computer deployment increases Triton demand — hardware-agnostic revenue model
- NQCC deployment: multiple Triton systems at UK National Quantum Computing Centre, Harwell
- Korea opportunity: Samsung and SK Hynix quantum computing programmes will require Triton systems — direct supply chain relationship
- FTSE All-Share: established public company with reliable revenue model from quantum infrastructure — less volatile than pure-play quantum stocks

17. QLM Technology

Company	QLM Technology Ltd
Founded	2018 — Bristol, UK (University of Bristol Quantum Technology Enterprise Centre spinout)
CEO	Murray Reed
Technology	Quantum gas lidar: single-photon detection cameras for methane and GHG emissions monitoring
Key Product	Quantum lidar camera: detects, locates, and quantifies methane leaks at industrial facilities, 200m range
Funding	£12M Series A (SLB/Schlumberger lead + Development Bank of Wales + Quantum Exponential Group)
Partnership	Project Canary (April 2025): expanding methane monitoring to US gas distribution companies
Market	Oil & gas pipelines, landfills, wastewater treatment, industrial facilities — emissions compliance

Technology & Strategic Position

QLM Technology applies quantum sensing to one of the world's most commercially urgent environmental problems: measuring methane leaks from industrial infrastructure. Its quantum lidar cameras use tunable diode lidar with single-photon detection to identify and quantify methane concentrations at up to 200 metres range — far more sensitive than classical infrared cameras and capable of detecting leaks invisible to conventional sensors.

SLB (formerly Schlumberger) — the world's largest oil services company — leading QLM's Series A and gaining access rights to QLM technology for its emissions monitoring business represents a definitive commercial validation. The Project Canary partnership (April 2025) expands QLM's reach to US gas distribution operators, where methane monitoring is now legally mandated under EPA regulations. For Korea, POSCO, HD Hyundai, and Korea Gas Corporation face equivalent emissions monitoring mandates under K-ETS.

- SLB (Schlumberger) Series A lead: world’s largest oil services company choosing QLM as preferred quantum emissions monitoring partner
- Project Canary partnership (April 2025): expanding methane monitoring capabilities to US gas distribution companies under EPA mandates
- UP-SENSE Innovate UK award: next-generation quantum gas sensors using photon up-conversion and SPAD detectors
- Korea emissions mandate: POSCO, HD Hyundai, Korea Gas Corporation require methane monitoring under K-ETS — direct QLM market

18. Quantum Dice

Company	Quantum Dice Ltd
Founded	2020 — Oxford, UK (University of Oxford spinout)
CEO	Ramy Shelbaya
Technology	QRNG hardware: source-device-independent quantum random number generation
Key Feature	Self-certifying: continuously verifies quantum nature of randomness in real-time (no need to trust the hardware)
Market	Cryptographic key generation, quantum-secure communications, Monte Carlo simulation, gaming
Funding	£5M+ early stage with UK government grants
Distinction	Only commercial QRNG device with source-device-independence — provably quantum random without trusting the device

Technology & Strategic Position

Quantum Dice’s QRNG technology generates random numbers from quantum physical processes (photon arrival times) and simultaneously verifies in real time that the randomness is genuinely quantum in origin — not from a compromised or faulty device. This source-device-independent (SDI) protocol solves the fundamental trust problem in cryptographic random number generation: users need not trust the hardware, because the hardware continuously proves its own quantum behaviour.

For Korean financial institutions (KEB Hana Bank, KB Financial, Samsung Securities) and government agencies migrating to NIST PQC standards, QRNG is a critical component: PQC algorithms are only as secure as the random numbers used in key generation. Quantum Dice’s SDI protocol provides the cryptographic assurance of genuinely random keys that government-grade security requires.

- SDI QRNG: self-certifying quantum randomness — device continuously proves it is generating genuine quantum random numbers
- Oxford spinout: world-class quantum optics physics base, proximity to Oxford quantum ecosystem
- NIST PQC integration: Quantum Dice QRNG provides the high-quality entropy source required for CRYSTALS-Kyber and CRYSTALS-Dilithium key generation

- Korea FSC relevance: Financial Services Commission quantum security guidelines requiring certified QRNG for high-assurance key generation

19. Post-Quantum

Company	Post-Quantum Ltd
Founded	2009 — London, UK (one of the first PQC companies globally)
CEO	Andersen Cheng
Technology	Post-quantum cryptography: identity verification, messaging, and network security
Key Products	PQ3 (post-quantum messaging), PQxDH (X3DH replacement), NIST lattice-based crypto implementations
Military	US DoD and NATO contracts for post-quantum secure communications
Standards	IETF standards contributions — Post-Quantum's protocols influence global PQC standards
Distinction	Only UK PQC company with active US DoD and NATO military contracts

Technology & Strategic Position

Post-Quantum was building quantum-resistant cryptography before the NIST PQC competition began — giving it a decade of head start on NIST-compliant implementations and military deployment. Its PQxDH (post-quantum X3DH) protocol replaces the key agreement mechanism used in WhatsApp, Signal, and other encrypted messaging applications with a quantum-safe equivalent. US DoD and NATO contracts represent the highest-value government quantum security engagements available.

For Korea, Post-Quantum's military PQC credentials are directly relevant to ROK defense requirements. The ROK Ministry of National Defense is actively planning cryptographic migration for classified systems — Post-Quantum's existing NATO relationship provides a credibility pathway into the Korean defense procurement process, particularly given AUKUS technology cooperation frameworks.

- US DoD contracts: quantum-safe communications systems deployed in US defense infrastructure
- NATO PQC implementations: post-quantum cryptographic protocols deployed in NATO alliance communications
- PQxDH protocol: post-quantum replacement for X3DH key agreement — influences next generation of end-to-end encrypted messaging
- IETF standards: Post-Quantum contributing to global internet standards for post-quantum key agreement
- Founded 2009: 16-year head start on NIST PQC — deeper implementation experience than any other UK PQC company

20. National Quantum Computing Centre (NQCC)

Organisation	National Quantum Computing Centre (NQCC) — UK Government, Harwell Campus, Oxfordshire
Founded	2021 — UK Research and Innovation (UKRI)
Director	Michael Cuthbert
Role	UK's sovereign quantum computing infrastructure hub and national user facility
Systems	IBM Quantum System Two, OQC Toshiko, Rigetti system, Google Willow cloud access
Target	1 Million Quantum Operations (1M-QuOp) by 2028 — UK national quantum benchmark
Funding	Part of £670M UK government quantum commitment · UKRI £76M compute initiative (March 2026)
Partners	IBM, OQC, Rigetti, Google, Riverlane, ORCA, Universal Quantum, UK universities

Strategic Significance

The NQCC is not a company, but it is arguably the most strategically important organisation in the UK quantum computing ecosystem. As the national user facility, it provides UK researchers, enterprises, and government agencies access to the full range of quantum hardware platforms — IBM, OQC, Rigetti, and Google Willow — under sovereign UK infrastructure that is not subject to US technology export controls. The 1M-QuOp target (one million reliable quantum operations by 2028) is the UK's primary national quantum computing benchmark.

For Korea, the NQCC represents a bilateral research partnership opportunity rather than a commercial relationship. Korea's KIST, ETRI, and KAIST quantum research programmes could establish NQCC access agreements that provide Korean researchers exposure to multiple quantum hardware platforms simultaneously — accelerating Korea's quantum workforce development without building domestic infrastructure from scratch.

- UK sovereign quantum infrastructure: multiple hardware platforms (IBM, OQC, Rigetti, Google Willow) under UK government operation
- 1M-QuOp 2028 target: national quantum benchmark driving QEC research, hardware deployment, and algorithm development
- UKRI £76M National Compute Resources initiative (March 2026): four new National Compute Resources including quantum access
- Riverlane QEC integration: NQCC serves as proving ground for Deltaflow QEC stack across multiple hardware platforms
- Korea partnership pathway: KIST/ETRI/KAIST bilateral access agreement would provide Korean researchers multi-platform quantum access under UK-ROK Science and Innovation Partnership

Implications for Korea Market Entry

UK-Korea Quantum Strategic Alignment

The United Kingdom and South Korea have established a Global Strategic Partnership (GSP) that explicitly covers science and technology, providing the bilateral framework for quantum technology cooperation. UK-Korea cooperation in quantum spans academic research (Cambridge-KAIST linkages, Oxford-Seoul National University partnerships), commercial enterprise (Quantinuum's enterprise customer model applicable to Korean conglomerates), and defense (QinetiQ-ROK MND through Five Eyes-adjacent cooperation).

Korea Need	UK Company	Strategic Opportunity
K-Moonshot QC (2028)	Quantinuum, OQC, ORCA	Hardware deployment · KAIST/KIST joint QC research
QEC Infrastructure	Riverlane Deltaflow	NQCC model replicable at Korean national QC centre
PQC Chip (IoT/Automotive)	PQShield, Crypto Quantique	Samsung Semiconductor, Hyundai Mobis PQC chip IP
Quantum-Safe Networks	KETS QKD + Arqit SKA	KT/SKT/LG U+ quantum-secure telco infrastructure
QC Networking	Nu Quantum	Korea National Quantum Network modular architecture
GHG Emissions Sensing	QLM Technology	POSCO, Korea Gas Corp K-ETS methane monitoring
Defense Quantum	QinetiQ + Post-Quantum	ROK MND PQC migration · GPS-denied sensing
QC Infrastructure	Oxford Instruments Triton	Samsung/SK Hynix QC lab dilution refrigerators
Quantum Drug Discovery	Phasecraft + Quantinuum InQuanto	Samsung Biologics, Boryung pharma QC algorithms
Quantum Control SW	QuantrolOx	KAIST/POSTECH quantum lab automation platform

US vs France vs UK: Comparative Positioning for Korea

Dimension	United States	France	United Kingdom
Hardware	IBM/Google/IonQ — world-leading all modalities	PASQAL deployed in Korea, Quandela, Alice & Bob	Quantinuum world-record fidelity, OQC data-centre native
Korea Invest	No dedicated commitment — cloud access only	\$52M PASQAL committed, LG Electronics equity partner	No committed capital yet — partnership framework only
QEC	Riverlane (Cambridge/US dual HQ) — global leader	QPerfect (Quobly partnership)	Riverlane UK HQ — hardware-agnostic, 60% QC

Dimension	United States	France	United Kingdom
			coverage
PQC	NIST standards (SandboxAQ, SEALSQ)	SEALSQ/IC'Alps PQC chips	PQShield chip IP (Arm investment), Arqit, Post-Quantum
Defense	Lockheed, Northrop, Booz Allen	Thales quantum sensing	QinetiQ defense quantum, Post-Quantum NATO/DoD
ITAR Risk	Tech export controls may restrict to Korea	No equivalent restriction	No ITAR equivalent — open bilateral technology transfer

Strategic Recommendation: The optimal Korea quantum strategy requires all three national ecosystems working in parallel, each filling a distinct role. The US provides dominant cloud platforms (AWS Braket Korea, Azure Quantum) and NIST PQC standards. France provides the only committed hardware capital in Korea (PASQAL \$52M) and the most transferable semiconductor manufacturing path (Quobly-Samsung Foundry). The UK provides the world's highest-fidelity hardware (Quantinuum) for enterprise applications, hardware-agnostic QEC infrastructure (Riverlane), PQC hardware IP at silicon level (PQShield for Samsung/Hyundai), and defense quantum cooperation through the UK-Korea Global Strategic Partnership. These three tracks are structurally complementary and should be pursued simultaneously.

Conclusion

The United Kingdom has built a quantum ecosystem that is narrow by headcount but formidable by quality. Quantinuum's world-record fidelity, Oxford Ionics' \$1.075B exit, Riverlane's 60%+ hardware partnership rate, ORCA's room-temperature deployability, Nu Quantum's quantum networking leadership, and PQShield's NIST-certified PQC silicon represent a portfolio of quantum capabilities that no other country—including the US and France—replicates exactly.

The UK government's £670M June 2025 commitment — motivated explicitly by concern over foreign acquisition of UK quantum assets — signals that sovereign industrial quantum capability is now a UK national security priority. This investment creates the infrastructure and incentive structure for UK quantum companies to scale domestically rather than sell to US acquirers, preserving the ecosystem for international partnership.

For Korea, the UK quantum ecosystem offers something unique: a partner with complementary strengths rather than competing ambitions. The UK is not trying to build a Korean quantum market presence — it is open to bilateral partnerships that leverage UK quantum expertise to accelerate Korea's K-Moonshot 2028 programme and long-term quantum sovereign capability. The UK-Korea Global Strategic Partnership, Cambridge-KAIST research linkages, and the AUKUS quantum cooperation framework provide the bilateral architecture for this partnership to deepen rapidly.

Report prepared by Paul Oh (오성일)

Executive Member, Korean Society for Naval Science & Technology | Member, Quantum Information Society of Korea | www.paulncompanies.com | os81paul@gmail.com

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This report was compiled from publicly available sources as of March 14, 2026. All company information reflects publicly disclosed data. This report does not constitute investment advice.